Current State: Conversational Copilot Development

Productivity is improving, but delivery remains user-driven, conversational, and difficult to govern.

HOW WORK HAPPENS TODAY

1. Developer prompt
2. Requirements discussed in chat
3. Agent makes design assumptions
4. Code generated iteratively
5. Re-prompt, patch, and retry
6. Review and validation happen later

VALUE TODAY

- Faster exploration and prototyping
- Quicker code and documentation drafts
- Low barrier to adoption
- Improved individual developer productivity

CURRENT LIMITATIONS

- Requirements and decisions stay hidden in conversations
- Architecture decisions can be implicit or inconsistent
- Context must be repeatedly supplied
- Agent behavior varies by user and prompt
- Limited traceability across requirement, design, and code
- More rework when assumptions are incorrect
- Governance occurs late in the process

Effective for individual assistance, but not yet a repeatable enterprise delivery model.

Shared Agent Foundation: Approved Agents from a Common Repository

Move from individual prompting to reusable, version-controlled, enterprise-approved agents.

Product teams

Application repos

SHARED AGENT REPOSITORY

- Approved agent roles
- Version-controlled instructions
- Tool and permission definitions
- Spec and design templates
- Domain and capability context
- Architecture guardrails
- Workflow templates and test cases

Engineering platforms

Developer environments / workflows

OPERATING MODEL

1. Define once
2. Review and approve
3. Version and test
4. Publish through catalog
5. Reuse across teams

BENEFITS

- Consistent agent behavior
- Less repeated prompting
- Reuse of approved patterns
- Faster onboarding
- Clear ownership & accountability
- More predictable outputs

Centrally governed, locally executed, and safely reused.

Architecture Control Plane: Event-Driven Assurance

Independently validate agent-produced artifacts at key lifecycle events as work progresses.

EVENTS AND INPUTS

- Requirement spec created
- Technical design updated
- Development started
- Pull request opened
- Test results submitted
- Release approval requested

CONTROL PLANE

1. Receive event
2. Retrieve artifact and context
3. Validate against standards
4. Produce findings and decision
5. Record evidence and return feedback

- Traceability
- Architecture patterns
- Security and data controls
- Platform standards
- Quality and NFRs
- Required approvals

CONTROL-PLANE DECISIONS

- Pass
- Warn
- Block
- Exception
- Rework

BUSINESS VALUE

- Independent assurance
- Early issue detection
- Better governance and audit trail
- Standardized feedback to agents
- Safer progression through the SDLC

Reactive, not mandated. Validates after the fact and only on emitted events — gaps remain between events and on ungoverned paths. **Preventive, gate-before-execute governance arrives in the Target State.**

Target State: Governed, Capability-Grounded Agentic SDLC

Specialized agents collaborate across the lifecycle using approved context, shared controls, and human accountability.

GOVERNANCE LAYERS

- Shared agent repository
- Approved specifications
- Control plane
- Human approvals

AGENTIC LIFECYCLE

Co-design = human leads and AI agents shaping requirements and design together — not solo prompting.

1. Discover and clarify
2. Co-design requirements with AI agents
3. Design the solution
4. Review and approve
5. Generate execution workflow
6. Execute development
7. Validate continuously
8. Human review and approval
9. Deploy and observe

CAPABILITY-GROUNDED CONTEXT

- Business capability & domain vocabulary
- Product & customer context
- Approved architecture patterns
- Platform & technology standards
- Security, risk & compliance
- Repository & codebase context

TARGET OUTCOMES

- Clear source of truth
- Lower unsupported assumptions
- Reduced prompting and rework
- Repeatable implementation
- Built-in traceability
- Predictable quality and compliance

Co-design with AI agents on approved context. Execute from an approved specification. Govern through evidence.

The Maturity Curve: From Conversational Copilot to Autonomous SDLC

Each stage compounds capability and governance — autonomy is the horizon, earned on top of mandated control.

TODAY

Conversational Copilot

Ad-hoc prompting; design and code in one chat.

UNGOVERNED

Shared Agent Foundation

Reusable, version-controlled, approved agents.

STANDARDIZED

Event-Driven Control Plane

Detective assurance; validates events after the fact.

REACTIVE

TARGET STATE

Governed Agentic SDLC

Co-design with agents + mandated gate-before-execute.

PREVENTIVE

FUTURE

Autonomous SDLC

Agents plan, build, verify & ship within policy; humans set intent.

SELF-GOVERNING

Autonomy is the destination — but it is earned on top of mandated governance: agents gain independence as verification and guardrails prove out.